

# Insup Lee

Cybersecurity & AI Researcher

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## Research Interests

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My primary research focuses on **AI for cybersecurity and threat intelligence**. I have developed **domain-specific generative models** that preserve structural properties downstream classifiers require, ranging from continuous physical signals in wireless communications to discrete TTP sequences in cyber threat analysis. I recently expanded this foundation to LLMs for cyber threat knowledge graph extraction, shifting my focus from synthetic data generation to automated intelligence extraction. To advance this trajectory, I intend to deploy **agentic AI systems** to automate complex security tasks and investigate their vulnerabilities, contributing to the field of **securing AI**.

## Education

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**Korea University** Seoul, Korea  
*Ph.D. in Information Security* Sep 2019 - Present

- Advisors: Sangjin Lee and Seokhie Hong
- Dissertation: Domain-Specific Generative Models for Data Augmentation in Cybersecurity

**Korea University** Seoul, Korea  
*B.E. in Cyber Defense* Mar 2014 - Feb 2018

## Work Experience

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**Korea University** Seoul, Korea  
*Lecturer* Sep 2025 - Feb 2026

- Taught graduate-level course “Computer Networks”

**Ministry of National Defense** Korea  
*Cyber Officer* Aug 2023 - May 2025

- Deployed to Abu Dhabi, UAE for joint research on AI-based security (Apr 2024 - Mar 2025)

**Agency for Defense Development (ADD)** Seoul, Korea  
*Researcher* Jul 2018 - Jul 2023

- Mentor: Changhee Choi

## Selected Publications

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- [TIFS’25] **I. Lee et al.**, “MuCamp: Generating Cyber Campaign Variants via TTP Synonym Replacement for Group Attribution”
- [TDSC’24] **I. Lee et al.**, “UniQGAN: Towards Improved Modulation Classification With Adversarial Robustness Using Scalable Generator Design”
- [CAL’26] **I. Lee et al.**, “LeakDiT: Diffusion Transformers for Trace-Augmented Side-Channel Analysis”

## Full Publications

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**International Publications** (C: Conference, J: Journal)

C4 *Mnemonic Hack: Recovering the Master Seed from Bitcoin Hardware Wallets via Side-Channel Attacks*  
J. Baek, G. Ahn, S. Park, D. Bae, G. Kim, **I. Lee**, S. Hong, and H. Kim  
ACM Conference on Computer and Communications Security (**CCS’26**), Nov. 2026.

- C3 *Exploiting Per-Core Leakage: Electromagnetic Side-Channel Monitoring of Multicore Architectures*  
D. Bae, S. Park, **I. Lee**, Y. Jung, K. Lee, H. Kim, and S. Hong  
ACM/IEEE Design Automation Conference (DAC'26), Jul. 2026.
- J8 *LeakDiT: Diffusion Transformers for Trace-Augmented Side-Channel Analysis*  
**I. Lee**, D. Bae, S. Hong, and S. Lee  
IEEE Computer Architecture Letters, Vol. 25, No. 1, pp. 5-8, Jan./Jun. 2026.
- J7 *Multi-Step LLM Pipeline for Enhancing TTP Extraction in Cyber Threat Intelligence*  
H. Kim, D. Lee, **I. Lee**, S. Lee, and S. Lee  
IEEE Access, Vol. 13, pp. 179696-179710, Oct. 2025.
- J6 *Enhancing Modulation Classification via Diffusion Transformers for Drone Video Signal Processing*  
**I. Lee**, K. Alteneiji, and M. Alghfeli  
IEEE Signal Processing Letters, Vol. 32, pp. 3325-3329, Aug. 2025.
- J5 *MuCamp: Generating Cyber Campaign Variants via TTP Synonym Replacement for Group Attribution*  
**I. Lee** and C. Choi  
IEEE Transactions on Information and Forensics Security (TIFS), Vol. 20, pp. 6162-6174, Jun. 2025.
- J4 *UniQGAN: Towards Improved Modulation Classification With Adversarial Robustness Using Scalable Generator Design*  
**I. Lee** and W. Lee  
IEEE Transactions on Dependable and Secure Computing (TDSC), Vol. 21, No. 2, pp. 732-745, Mar./Apr. 2024.
- J3 *Camp2Vec: Embedding Cyber Campaign With ATT&CK Framework for Attack Group Analysis*  
**I. Lee** and C. Choi  
ICT Express, Vol. 9, No. 6, pp. 1065-1070, Dec. 2023.
- J2 *Exploiting TTP Co-occurrence via GloVe-Based Embedding With ATT&CK Framework*  
C. Shin, **I. Lee**, and C. Choi  
IEEE Access, Vol. 11, pp. 100823-100831, Sep. 2023.
- J1 *BAN: Predicting APT Attack Based on Bayesian Network With MITRE ATT&CK Framework*  
Y. Kim, **I. Lee**, H. Kwon, G. Lee, and J. Yoon  
IEEE Access, Vol. 11, pp. 91949-94968, Aug. 2023.
- C2 *Anomaly Dataset Augmentation Using Sequence Generative Models*  
S. Shin, **I. Lee**, and C. Choi  
IEEE International Conference on Machine Learning and Applications (ICMLA'19), Dec. 2019.
- C1 *Opcode Sequence Amplifier Using Sequence Generative Adversarial Networks*  
C. Choi, S. Shin, and **I. Lee**  
International Conference on ICT Convergence (ICTC'19), Oct. 2019.

### Domestic Publications (Korean)

(†: Corresponding Author)

- D3 S. Park, D. Bae, **I. Lee**, H. Kim, and S. Hong, "EM-Based Anomaly Detection using a Dual-Domain Approach," in Proc. of the KIISC Winter Conference (CISC-W), Nov. 2025. (Selected as an Outstanding Paper Award)
- D2 H. Park and **I. Lee**<sup>†</sup>, "Enhanced DDoS Detection via Traffic Volume-Based Labeling and Transfer Learning," Journal of Internet Computing and Services (JICS), Vol. 26, No. 4, pp. 1-8, Aug. 2025.
- D1 K. Kim and **I. Lee**<sup>†</sup>, "User Behavior Embedding via TF-IDF-BVC for Web Shell Detection," Journal of The Korea Institute of Information Security & Cryptology (JKIISC), Vol. 34, No. 6, pp. 1231-1238, Dec. 2024.

### Patents

- C. Choi and **I. Lee**, "Method for Augmentating Cyber Attack Campaign Data to Identify Attack Group, and Security," Korea Patent Application Number. 10-2024-0176082, December 2, 2024.
- C. Choi, **I. Lee**, C. Shin, and S. Lee, "Information Identification Method and Electronic Apparatus Thereof," Korea Patent Application Number. 10-2024-0006106, January 15, 2024.
- C. Choi, C. Shin, S. Shin, S. Seo, and **I. Lee**, "Method for Training Attack Prediction Model and Device Therefor," U.S. Patent Application Number. 18/126,005; U.S. Patent Number. US20230308462A1, September 28, 2023.
- C. Choi, S. Shin, and **I. Lee**, "Apparatus, Method, Computer-readable Storage Medium and Computer Program for Generating Operation Code," Korea Patent Application Number. 10-2019-0141865, November 07, 2019; Korea

Patent Number. 10-2246797, April 30, 2021.

## Other Experience

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|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|
| AI Cyber Challenge (AIXCC), DARPA and ARPA-H, USA                                                                                                                                                                                             | Apr 2024 – Aug 2024 |
| <ul style="list-style-type: none"><li>Participated in the semifinal round as a member of Team KORIA, submitting our cyber reasoning system that leverages LLMs for automated detection and patching of software vulnerabilities</li></ul>     |                     |
| SW Outsourcing Development, KCMVP-Certified Cryptographic Module                                                                                                                                                                              | Jun 2017 – May 2018 |
| <ul style="list-style-type: none"><li>Implemented a cryptographic module with 25,000 LoC in C - ARIA block cipher (modes: ECB, CBC, CTR), hash functions (SHA-256, SHA-512), and HMAC-based DRBG for Windows (.dll) and Linux (.so)</li></ul> |                     |

## Awards and Honors

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| KU Graduate School Achievement Award<br>Korea University, Seoul, Republic of Korea                   | Feb 2026            |
| Outstanding Paper Award<br>CISC-W'25, KIISC                                                          | Nov 2025            |
| Certificate of Commendation (UAE-ROK Engagement Program)<br>United Arab Emirates Ministry of Defense | Mar 2025            |
| Ambassador's Commendation<br>Embassy of the Republic of Korea to the United Arab Emirates            | Mar 2025            |
| Full Tuition Scholarship (Korea University)<br>Ministry of National Defense, Republic of Korea       | Mar 2014 – Feb 2018 |

## Professional Service

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### Reviewer

- IEEE Transactions on Dependable and Secure Computing (TDSC), 2025
- IEEE Transactions on Information Forensics and Security (TIFS), 2026
- IEEE Transaction on Communications (TCOM), 2025, 2026
- IEEE Journal on Selected Areas in Communications (JSAC), 2025, 2026

## Teaching Experience

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|-------------------------------------------------------------------------|---------------------|
| Lecturer, Computer Networks (SCS302, Fall 2025), Korea University       | Sep 2025 - Dec 2025 |
| Instructor, Penetration Testing - Intermediate, UAE Ministry of Defense | Sep 2024            |
| Instructor, Penetration Testing - Basic, UAE Ministry of Defense        | Jul 2024            |

## Mentoring Experience

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| <b>Sujin Park</b> (Ph.D. Student at Korea University)<br>Project: Side-channel analysis for anomaly detection [D3]                 | Jun 2025 – Present  |
| <b>Hyoungrok Kim</b> (M.S. Student at Korea University)<br>Project: LLM-based TTP extraction for cyber threat intelligence [J7]    | Mar 2025 – Sep 2025 |
| <b>Hyunjun Park</b> (Navy Lieutenant at Ministry of National Defense)<br>Project: DDoS detection via transfer learning [D2]        | Nov 2024 – Feb 2025 |
| <b>Kangmun Kim</b> (First Lieutenant at Cyber Operations Command)<br>Project: Web shell detection via user behavior embedding [D1] | Jan 2024 – Sep 2024 |

## Reference

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**Sangjin Lee**

*Professor*, School of Cybersecurity, Korea University

Relationship: Ph.D. Advisor

Email: [sangjin@korea.ac.kr](mailto:sangjin@korea.ac.kr)

**Seokhie Hong**

*Former Professor*, School of Cybersecurity, Korea University

Relationship: Ph.D. Advisor

Email: [shhong@korea.ac.kr](mailto:shhong@korea.ac.kr)

**Changhee Choi**

*Associate Professor*, Department of Cyber Defense, Sejong University

Relationship: Research Mentor (Former PI at ADD)

Email: [choich@sejong.ac.kr](mailto:choich@sejong.ac.kr)

**Jongin Lim**

*Professor Emeritus*, School of Cybersecurity, Korea University

Former Special Presidential Advisor for Cybersecurity, Republic of Korea

Relationship: Undergraduate Advisor

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